

CYL1010- Chemistry
Major Examination, Semester I, AY 2025-26
Department of Chemistry
Indian Institute of Technology Jodhpur

- Note: 1. All questions are compulsory
2. Understanding the question for a solution is also part of examination
3. All parts of each question should be answered together.
4. Questions in each section should be preferably attempted together.

Time: 2-5 PM

Total Marks: 100

Section A

[Total Marks: 10]

1. (a) For a Harmonic Oscillator the raising and lowering operators can be defined as $a_+ = \left(\frac{m\omega}{2\hbar}\right)^{1/2} x - i\left(\frac{1}{2m\omega\hbar}\right)^{1/2} p$ and $a_- = \left(\frac{m\omega}{2\hbar}\right)^{1/2} x + i\left(\frac{1}{2m\omega\hbar}\right)^{1/2} p$, respectively. Show that
 - (i) $[a_-, H] = \hbar\omega a_-$ where H is the Hamiltonian operator
 - (ii) Use $[a_+, H] = -\hbar\omega a_+$ and the result in (i) to show that $Ha_- \psi_n = (E - \hbar\omega) a_- \psi_n$
- (b) Describe the effect of conjugation on bathochromic shift using the molecular orbital diagram for ethylene, 1,3-butadiene, and 1,3,5-hexatriene [3+2]
2. (a) Which of the following statements are correct [2+2+1]
 - (i) All solutions of Schrodinger equation (ψ_n) for Harmonic oscillator are eigen functions of $a_+^2 a_-^2$
 - (ii) The selection rules for 1-D Harmonic oscillator allow for $\Delta n = \pm 2$
 - (iii) Polar solvents result in hypsochromic shift in $n \rightarrow \pi^*$ absorption band of α, β - unsaturated carbonyl compounds
 - (iv) The wave functions for 1-D Harmonic oscillator are always symmetric
- (b) Justify whether the following statements are correct or not (**No justification carries zero mark**)
 - (i) For a 2-dimensional Harmonic oscillator, the wave function is represented as ψ_{n_x, n_y} where n_x and n_y are respective quantum numbers. If ψ_{12} and ψ_{21} are eigen functions of the Hamiltonian operator, then $c_1\psi_{12} + c_2\psi_{21}$ will also be an eigen function of the Hamiltonian operator
 - (ii) CO_2 is IR inactive as the permanent dipole moment for the molecule is 0

Section B

[Total Marks: 40]

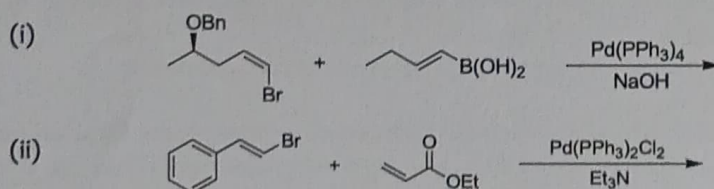
3. Given bulk band gaps of CdS (512 nm) and CdSe (716 nm), CdSe show more colours than CdS. Explain it using the Brus equation [3]
4. How does surface area to volume ratio change for a cubical nanoparticle when its size decreases from 1000 nm $\rightarrow$ 100 nm $\rightarrow$ 10 nm? Draw a plot. [3]
5. In Fermi-Dirac distribution:
 - a) What is $f(E)$ at $E = E_F$ for any $T > 0$? [1]
 - b) Between temperatures T_1 and T_2 ($T_2 > T_1$), which shows a sharper drop in $f(E)$? Explain. [2+2 = 4]

6. Show mathematically that the surface tension of a liquid is equal to the surface energy per unit area. A capillary of diameter 0.21 mm is dipped in a liquid of density 0.79 g/cm³. The liquid rises 6.30 cm. Take $g = 980 \text{ cm/s}^2$. Calculate the surface tension. Why does surface tension decrease with temperature? [2+3+2 = 7]
7. Using BET theory: Calculate the specific surface area given $V_m = 50 \text{ cm}^3/\text{g}$ and $a = 0.162 \text{ nm}^2$. [3]
8. How many grams of copper are deposited on a cathode if 2.00 A current flows through CuSO₄ solution for 20 min? [3]
9. Explain how Kronig–Penney model help in understanding electronic band structures of insulators, semiconductors and metals. [5]
10. Why do Zinc–Carbon batteries commonly show casing corrosion and leakage? [3]
11. List four advantages of using KOH in alkaline batteries over NH₄Cl in Zinc–Carbon batteries. [4]
12. Write the reversible chemical reaction of:
 - a) Lead–acid battery [2]
 - b) Lithium-ion battery [2]

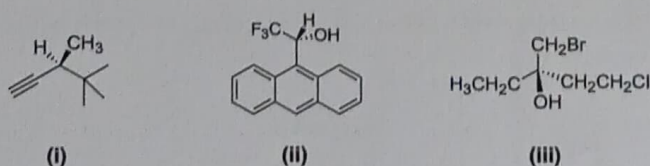
Section C

[Total Marks: 50]

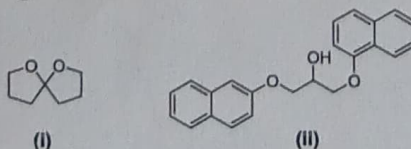
13. Identify the cross-coupling reaction given below and write the step-by-step catalytic cycle and the corresponding product for each: (2 x 4 = 8 Marks)



14. Indicate whether each of the following structures has the R or the S configuration: (3 x 1 = 3 Marks)

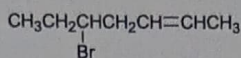


15. A solution prepared by mixing 30 mL of a 0.20 M solution of the R enantiomer and 20 mL of a 0.10 M solution of the S enantiomer was found to have an observed specific rotation of $+7.8^\circ (\text{g/ml})^{-1} \text{ dm}^{-1}$. What is the specific rotation of each of the enantiomers? (2 Marks)
16. Justify whether the following molecules are chiral or achiral: (2 x 1 = 2 Marks)

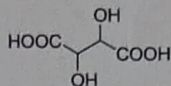


17. Draw all the possible stereoisomers for the following compounds:

(2 x 1.5 = 3 Marks)

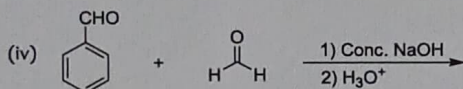
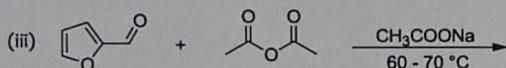
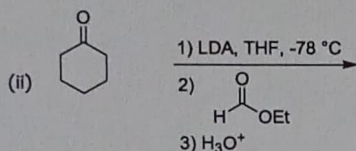
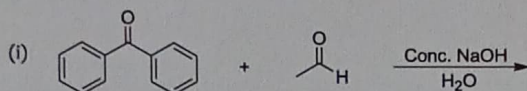


(i)



(ii)

18. Write the possible product/s with the plausible mechanism for the following reaction: (4 x 4 = 16 Marks)



19. Discuss the cooperativity in haemoglobin's oxygen binding, referring to the structural differences between oxyhaemoglobin and deoxyhaemoglobin. Draw the oxygen-binding curve for haemoglobin. [4 Marks]

20. Justify the following by writing electronic configuration of the transition metal ion in the complex, drawing the crystal field splitting diagrams and considering various selection rules: [3 x 3 = 9 Marks]

(i) The complex $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is light pink in color, whereas, the complex $[\text{CoCl}_4]^{2-}$ shows deep blue color.

(ii) The green color of complex $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ turns ^{violet} when ammonia is added to give $[\text{Ni}(\text{NH}_3)_6]^{2+}$.

(iii) The complexes $[\text{Cu}(\text{PPh}_3)_4]^+$, $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cr}(\text{CO})_6]$ are colorless.

21. A buffer solution is prepared by mixing 0.40 moles of acetic acid (CH_3COOH) and 0.30 moles of sodium acetate (CH_3COONa) in enough water to make 1.0 L of solution. Given that the pK_a of acetic acid is 4.76, calculate: [3 x 1 = 3 Marks]

(i) The pH of the buffer solution.

(ii) The pH after adding 0.01 moles of HCl to the buffer (assume no volume change).

(iii) The pH after adding 0.01 moles of NaOH to the buffer (assume no volume change).

Total Marks: 50, Time: 3 hrs, Date: 27. 4. 2025

Physical Section (30 Marks)+Inorganic Section (15 Marks) + Organic Section (5 Marks)

There are a total 5 pages in the question paper.

- $c = 2.9979 \times 10^{10} \text{ cm/s}$.
- $1\text{N} = \text{kg.m/s}^2$
- $\pi = 3.141$
- $h = 6.626 \times 10^{-34} \text{ Js}$.

1. The wave function of an electron in the lowest energy state of a hydrogen atom is proportional to e^{-r/a_0} with a_0 a constant and r as the distance from the nucleus. By what factor probability of finding the electron inside a region of volume $\delta V = 1.0 \text{ pm}^3$, which is small even on the scale of the atom will be located at the nucleus than a distance a_0 from the nucleus?

point 2

2. The lowest energy electrons of a carbon nanotube can be described by the normalized wavefunction $(\frac{2}{L})^{1/2} \sin(\pi x/L)$, where L is the length of the nanotube. What is the percentage of chance to find the electron between $x = L/4$ and $x = L/2$?

point 2

3. a) $\psi_{200} = (\frac{1}{\sqrt{32\pi}})(\frac{1}{a_0})^{3/2}(2 - \frac{r}{a_0})e^{-r/(2a_0)}$ for the hydrogen atom. Evaluate the location of the node.
b) One of the p orbital has a wave function $(\frac{3}{4\pi})^{1/2} \sin\theta \cos\phi$. Identify which p orbital is this?

point 1+1

4. An electron is accelerated from rest through a potential difference of 40 kV.

a) What is its final deBroglie wavelength in pm? Charge of an electron = $1.609 \times 10^{-19} \text{ C}$, mass of $e = 9.109 \times 10^{-31} \text{ kg}$, $h = 6.626 \times 10^{-34} \text{ Js}$, $1\text{VC} = 1 \text{ J}$, $\text{J} = \text{kgm}^2\text{s}^{-2}$.

b) From the answer of the above problem, can you predict if electron diffraction can be suitable to visualize biological systems where typical bondlengths in molecules are about 100 pm? Justify your answer.

point 2+2

5. Write down with justification if the following functions are acceptable.

a) $f(x) = |x|$,

b) $\phi(x) = \frac{e^{-x^2}}{1+e^{1/x}}$

point 2+2

6. Can you identify the light emitted when a particle of mass 1×10^{-27} g in a $3 \text{ \AA}$ one dimensional box goes from the ground to the 1st excited state using UV-VIS spectroscopy? Mass of the particle is 10^{-30} kg. The range of UV is 100 – 400 nm, visible range is 400 – 700 nm, IR range is 780 nm-1 mm. Justify your answer.

point 2

7. The two possible wavefunctions for an electron confined to a one-dimensional quantum dot (a collection of atoms with dimensions in the range of nm) are $\sin x$ and $\sin 2x$. a) Can these wave functions have eigenvalues of kinetic energy operator? Justify your answer.

b) When the electron with wavefunction $\sin x$ is excited to higher energy states, its wave function becomes $\sin 3x$. Are these two wavefunctions orthogonal to each other? You can take the range from 0 to 2π . Justify your answer.

point 2+2

8. Suppose the probability distribution $f(x) = \frac{1}{b-a}, a \leq x \leq b$, $f(x) = 0$ otherwise. Calculate $\langle x \rangle$.

point 2

9. a) Identify the basic property on which the principle of MRI depends on: i) tunneling, ii) nuclear spin, iii) transition between two electronic states, iv) electronic spin.

b) Identify one basic property based on which atomic clock and further GPS are built up: i) Rabi oscillations, ii) heisenberg's uncertainty principle, iii) wave particle duality.

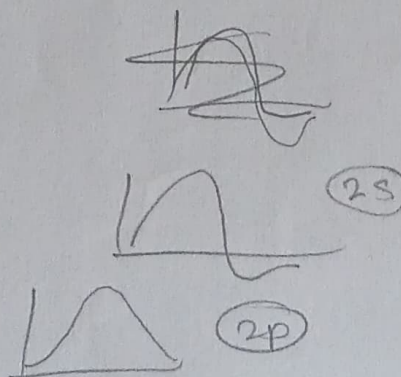
point 1+1

10. $\psi_{2s} = (\frac{1}{\sqrt{32\pi}})(\frac{1}{a_0})^{3/2}(2 - \frac{r}{a_0})e^{-r/(2a_0)}$. $\psi_{2p} = (\frac{1}{\sqrt{32\pi}})(\frac{1}{a_0})^{3/2}(\frac{r}{a_0})e^{-r/(2a_0)}\cos\theta$. Between 2s and 2p orbital of the hydrogen atom, which orbital is smaller in size? Justify your answer by plotting radial distribution function.

point 2

11. Calculate the uncertainty in speed if we wish to locate an electron within an atom, say, so that Δx is approximately 50 pm. Mass of an electron is 9.11×10^{-31} kg.

point 1



12. Write down the ground state configuration of the O^{2-} ion.

point 1

13. a) Write down the form of the wave function of He atom in terms of Hartree orbitals.

b) What quantum numbers are needed to denote an orbital?

point 1+1

CYL1010
Inorganic section
Major exam (15 marks)

1. Identify the geometry and shape of the following molecules using VSEPR model (2 marks)

a) ClO_2^- b) $[\text{SO}_3]^{2-}$ c) $[\text{TeF}_5]^-$ d) XeOF_2

2. What are the point groups for the following molecules? (2 marks)

a) CO_3^{2-} b) *trans*- $\text{Cl(H)C}=\text{C(H)Cl}$ c) HOCl d) *mer*- $\text{M(CO)}_3\text{X}_3$

3. From the given list of ν_{CO} bands, match the correct values to the third row (5d) transition metal carbonyls. Justify your answer (2)

$[\text{Os(CO)}_6]^{2+}$, $[\text{Ta(CO)}_6]^-$, $[\text{Ir(CO)}_6]^{3+}$, $[\text{W(CO)}_6]$, $[\text{Hf(CO)}_6]^{2-}$, $[\text{Re(CO)}_6]^+$

1220, 2085, 2254, 2190, 1850, 1757

4. The intensity of a monochromatic visible light is reduced by 90% due to absorption on passing through a 5.0 mM solution of a compound. If the path length is 4 cm, then calculate the molar extinction coefficient of the compound. (2)

5. Given $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34$ volt; $E^\circ_{\text{Fe}^{2+}/\text{Fe}} = -0.44$ volt

Predict a spontaneous reaction and also construct an electrochemical cell for that reaction. (2 marks)

6. The first order reflection from a crystal plane in a cubic crystal system occurs at $13^\circ 41'$. Find the Miller indices of the plane given $a = 5.63 \text{ \AA}$ and $\lambda = 1.54 \text{ \AA}$. [$\sin^2 13^\circ 41' = 0.056$] (2 marks)

7. Arrange the given metal complexes in the increasing order of intensity of color (ϵ) shown by them. Justify your order by writing below each the status of the selection rules for these complexes (2 marks)

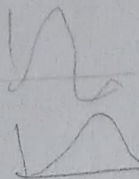
$\text{Fe}_4[\text{Fe(CN)}_6]_3$, $[\text{CoBr}_4]^{2-}$, $[\text{MnF}_6]^{4-}$, $[\text{Co(H}_2\text{O)}_6]^{2+}$

8. Arrange the following species in the order of increasing Jahn Teller distortion in an octahedral environment: (1 mark)

a) High-spin Co(III) b) Low-spin Co(II) c) Copper (II) d) Ti(III)

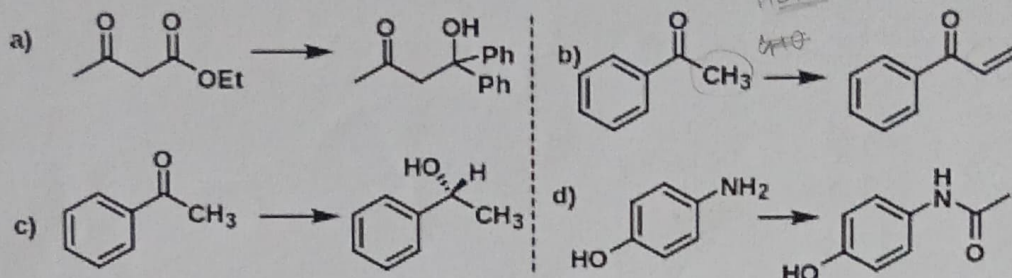
Sc Ti V Cr Mn Fe Co Cu

$A = \epsilon c l$



Organic Section [5 Marks]

1. How will you synthesis the following target molecules from given starting material? Write appropriate reagents and reaction-conditions stepwise (Mechanism not required). [4 Marks]



2. Find out the reagents/reactant A and products B and complete the following synthesis. (Mechanism not required). [1 Mark]

